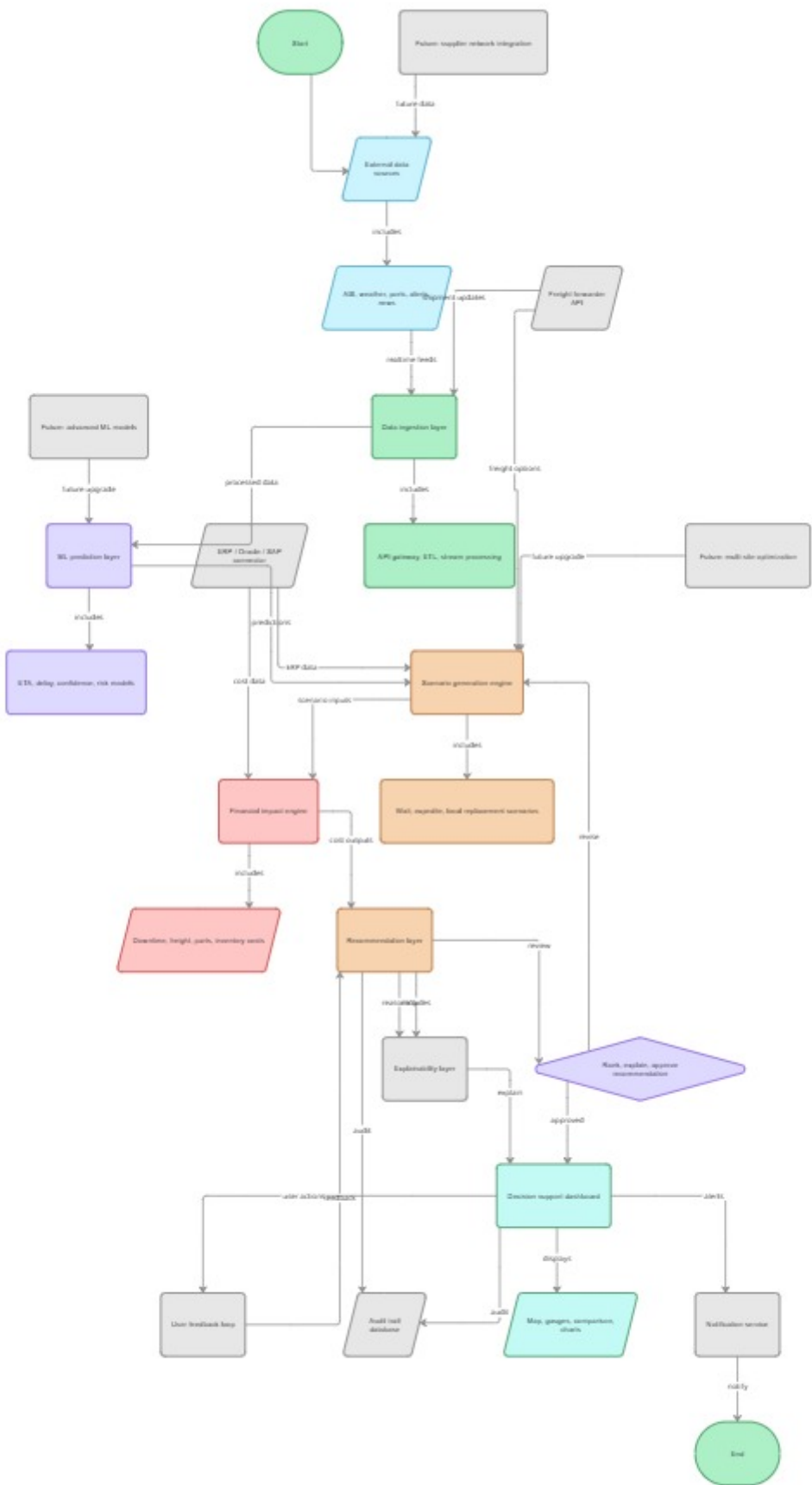








DenkKern Lena 2.0 —
Living Product System
Map

Comprehensive strategy
board for MVP validation
and execution

SECTION 1: Vision & Problem

From delay prediction to operational
decision support

- Not building a shipment tracking dashboard
- Help users answer: "What should we do now?"
- Convert shipment uncertainty into financial decision support

RISK: If we only show ETA and alerts, we become a fancy alerting tool

SECTION 2: Lena 2.0 Use Case

User journey:

Alert → Dashboard → Compare Scenarios → Review Recommendation → Decide → Approve

Context:

- Lena: Production manager, Hamburg shipbuilding
- Marine bolts arriving late
- Cost per delayed day: €150,000
- Choices: wait, expedite, order replacement parts

INSIGHT: Decision is about minimizing business loss, not controlling the ship

SECTION 3: Product System Flow

1. **External Signals:** AIS data, Weather, Port congestion, Maritime incidents, Strike risk, News
2. **Prediction Layer:** ETA, Delay probability, Confidence interval, Risk score
3. **Scenario Engine:** Wait | Expedite | Order replacement parts
4. **Financial Impact:** Downtime cost, Action cost, Expected loss, Total cost
5. **Decision Support:** Rank options, Explain trade-offs, Show reasoning
6. **Dashboard:** Scenario cards, Risk indicators, Recommendations, Timeline
7. **Human Decision:** Review, Select, Record

KEY: System ranks and explains. Lena decides.

Visualize a Technical Architecture — MVP

MVP Components (Build Phase)

- Data processing engine
- Prediction algorithms
- Manual data input interface (for managers)
- External input connectors
- Scenario calculator
- Reporter

Future Components (Post MVP)

- Real-time integration
- User interface
- Advanced scenario generation
- Automated workflows
- Knowledge layer
- Recommendation personalization

WARNING: Do not confuse these components with actual hardware

SECTION 5: Open Questions

- QO-001: Can customers quantify cost-of-delay defensibly?
QO-002: Who owns the cost-of-delay number?
QO-003: Can we calibrate cost model in 1-2 weeks?
QO-004: Is port congestion alone painful enough?
QO-005: Can customers accept manual data intake?
QO-006: What actions do managers actually value?
QO-007: What explainability level is required?
QO-008: Who is the economic buyer?
QO-009: What data are customers unwilling to share?
QO-010: What financial outcome justifies payment?

✓ **ACTION:** Nick validates through customer interviews

SECTION 6: Risks

- R-001: Cost-of-delay numbers may be subjective
R-002: Product becomes prediction dashboard, not decision support
R-003: Customers reject manual data intake
R-004: ERP integration too heavy for MVP
R-005: Prediction uncertainty reduces trust
R-006: Security concerns block pilots
R-007: Team overbuilds before validation

CRITICAL RISK:

Overbuilding before validation

Mitigation: Validate with customers first, build second

SECTION 7: Customer Insights

Interview Synthesis Template:

- Interview ID
- Customer role
- Pain observed
- Current workflow
- Existing tools
- Cost-of-delay estimate
- Decision options considered
- Willingness to pay
- Follow-up action

PROCESS: Every customer interview feeds this section. Build evidence incrementally.

Visualize a Technical Architecture — Future

MVP Service — Build Phase

- Data processing engine
- Prediction algorithms
- Manual data input interface (for managers)
- External input connectors
- Scenario calculator
- Reporter

Future Service — After Validation

- Real-time integration
- User interface
- Advanced scenario generation
- Automated workflows
- Knowledge layer
- Recommendation personalization

WARNING: Do not confuse these components with actual hardware







→ Open Questions drive weekly validation sprints
Linear pain delayed

 REVIEW:
See ranked options

 DECIDE:
Choose least costly action

 APPROVE:
Execute decision

 PRODUCTION
CONTINUES:
Loss avoided

→ MVP vs Future determines tech stack

→ Alert triggers Disruption detection

→ Scenarios connect to Recommendation logic

→ Customer pain drives Financial Impact priority

→ Decision logged for Evidence loop

→ OQ-001 feeds Financial Impact Engine

✓ Evidence: 2 customer interviews confirm €150k/day pain

→ OQ-003 validates Manual Intake process as R

✓ Validated: Manual intake acceptable for pilot

→ OQ-007 informs Scenario Engine decision

 Testing:
Cost model in real data

 MVP DISCONTINUED
R&D doesn't validate
"You are predicting cost of delay accurately?" → NO! IN SCOPES

→ R-004 enforces MVP Scope Limitation

 Next:
Scenario ranking with Lema feedback

→ R-006 drives Audit Trail requirement

